

Obstacle Avoiding Car

Title of the Project



VIT
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

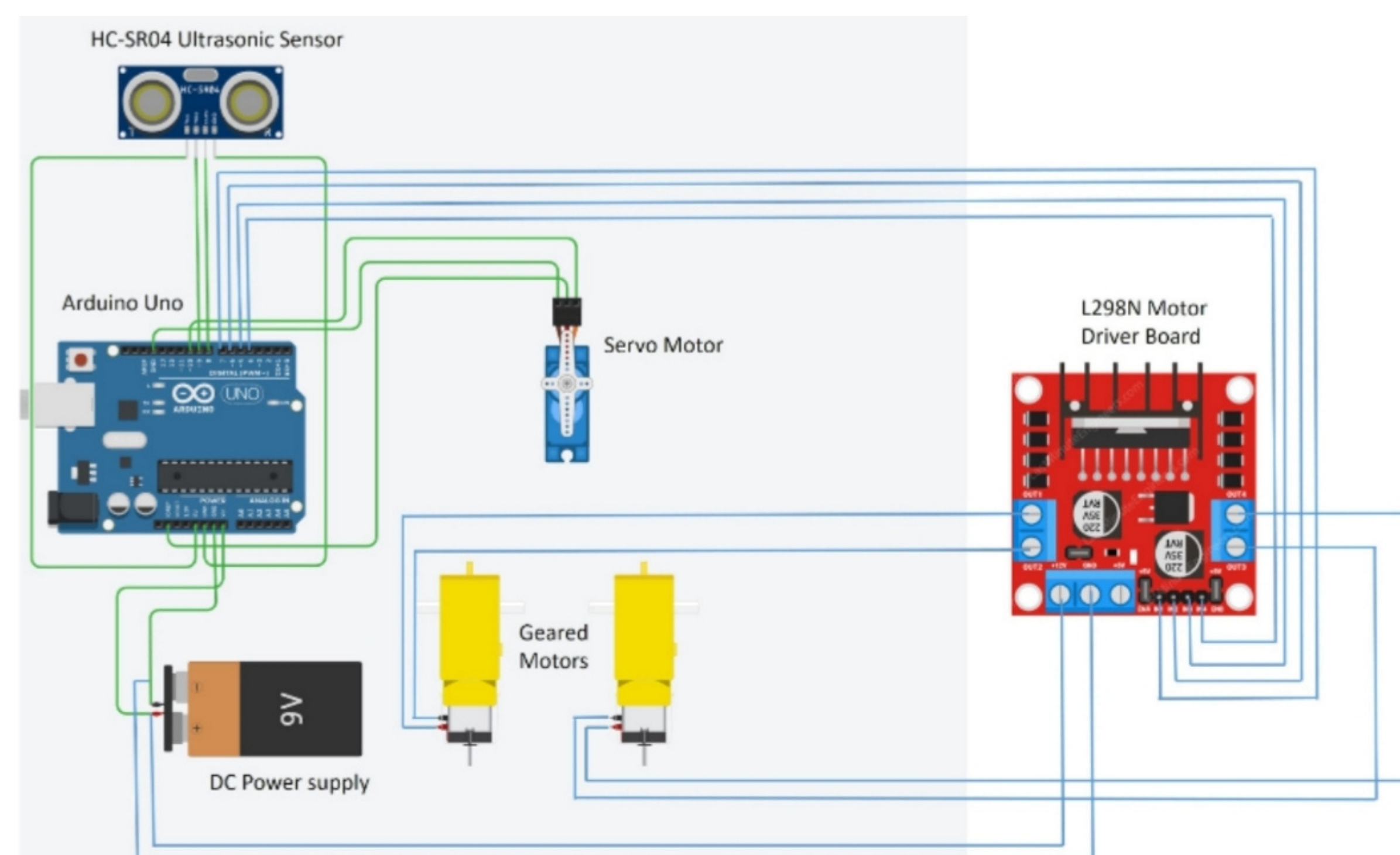
ObsCruiser

Objective

To create an obstacle avoiding car using Arduino that showcases robotics and automation in the field of engineering. The main aim is to integrate ultrasonic sensors, to code and to control systems to enable the vehicle to navigate through environments while avoiding obstacles .

The project aims to enhance road safety by preventing collisions with obstacles and reduce accidents. This initiative aligns with the broader goal of using technology to create solutions that improve daily life and safety standards.

Block Diagram



Arduino UNO Board - The Brain:

Acts as the brain of the car, processing input from sensors and controlling the motors

Motor Driver and Motors - The Driving Force:

Facilitates motor control, the motors navigate the car—forward, backward, and with nimble turns.

Ultrasonic Sensor - The Watchful Eye:

Using ultrasonic waves, this sensor spots obstacles, providing vital data for the avoidance system.

Servo Motor - The Direction Shifter:

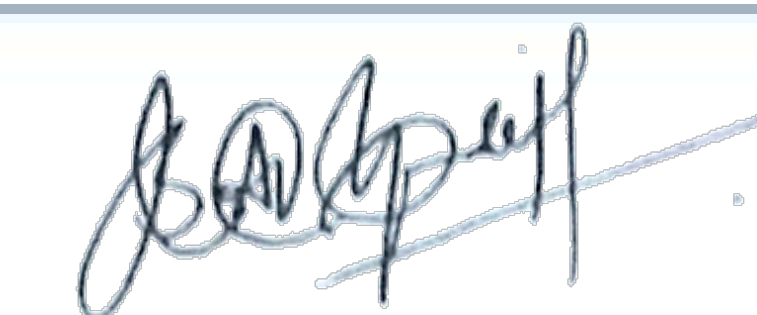
Controls the direction of the ultrasonic sensor, this motor ensures the car surveys its surroundings efficiently.

Social Relevance

Our project aims to lay the foundation for broader societal application like smart walking aids which are vehicles to assist visually impaired individuals in navigating urban environments safely, for search and rescue operations and enhancing unmanned



aerial vehicles (UAVs) or ground robots in negotiating challenging terrains. It could also be employed for surveillance and security purposes, warehouse automation , etc. In summary, our project holds the potential for a wide range of applications that could significantly benefit humanity.



Mentor Name - Dr. A. B. Ahadit

Team Members Reg. No. & Name

SCOPE:

23BAI1123 - Sk Abdul Khader
23BAI1157 - Anora V
23BAI1177 - Cherish Vasant
23BAI1194 - Lokeshwer Vijayakumar
23BAI1211 - Elena Margaret Ellis
23BAI1481 - Nishant ES

SENSE:

23BEC1009 - Lokesh.R
23BLC1264 - Samyukta Srikanth

SMEC:

23BMV1017 - S Akshatha
23BMV1025 - Kevin Edward J